CENG 252: Chemical Reaction Engineering

Study Notes for Lectures 6–10

Nonelementary Kinetics • Enzymes • Catalysis • Catalyst Deactivation

Based on the lecture slides (Z. Chen) and Fogler, *Elements of Chemical Reaction Engineering* (3rd ed. Ch. 7; 4th/5th ed. Ch. 10)

How to use these notes

- Each section starts with the **idea in one sentence**, then the **derivation** written out step by step (no skipped algebra), then a **boxed key result**.
- **Red boxes** are common mistakes, plus a few typos on the lecture slides. Read them: they are where exam points are lost.
- **Green boxes** are exam tips: which shortcut to use, and what the grader looks for.
- The last section is a **2-page formula summary**. The exam is closed-book with a formula sheet provided, but you must know *which* equation to use and how to set it up. Practise the derivations, not memorisation.

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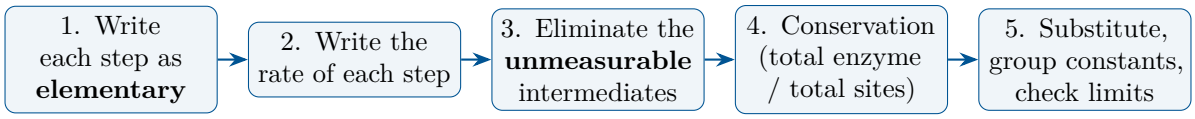
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1 The Big Picture: One Method for Every Mechanism

Lectures 6–9 all answer the same question:

Given a proposed mechanism of elementary steps, what rate law does it predict, and does that rate law match experiment?

The method is always the same, whether the intermediate is a free radical (Lec. 6–7), an enzyme–substrate complex (Lec. 7), or a species adsorbed on a catalyst surface (Lec. 8–9):



	Free radicals / active intermediates	Enzymes	Heterogeneous catalysis
Intermediate	A^*, H•, Br•, NO_3^*	E·S, E·I, I·E·S	A·S, B·S, …
How we eliminate it	PSSH: net rate of formation = 0	PSSH on E·S (or equilibrium of fast steps)	Non-limiting steps are at **equilibrium** ($r/k \approx 0$)
Conservation law	none needed	$C_{E,t} = C_E + C_{E\cdot S} + \dots$	$C_t = C_v + \sum C_{i\cdot S}$ (site balance)
Typical result	$\dfrac{kC_A^2}{1 + k'C_A}$	$\dfrac{V_{\max}C_S}{K_M + C_S}$	$\dfrac{kP_A}{1 + K_AP_A + K_BP_B}$

Reading a rate law: rules of thumb (Fogler Ch. 7 and Ch. 10)

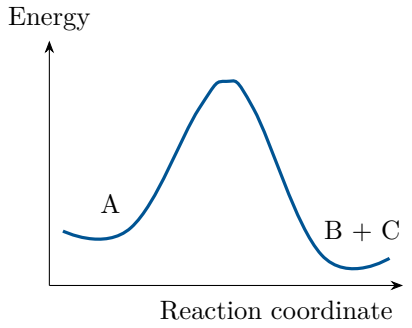
1. A species in the **denominator** collides with (or is adsorbed next to) the intermediate.
2. A **constant "1"** in the denominator means one step is a spontaneous decomposition of the intermediate (homogeneous kinetics), or stands for the vacant sites (catalysis).
3. A **fractional order** ($\frac{1}{2}$, $\frac{3}{2}$) points to a molecule splitting into two radicals or atoms (chain initiation, or dissociative adsorption).
4. A **squared denominator** $(1 + \dots)^2$ means a dual-site surface reaction (two sites are needed).
5. A species in the **numerator but not the denominator** reacts from the gas phase (Eley–Rideal) or is not adsorbed.

2 Lecture 6: Active Intermediates and the PSSH

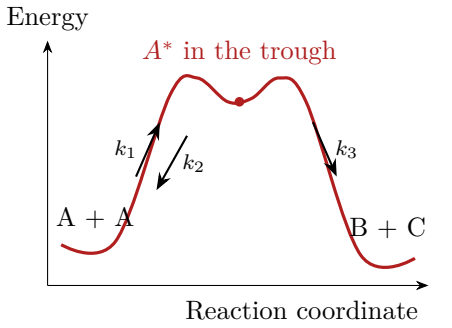
Fogler: 3rd ed. §7.1

2.1 What is an active intermediate?

An **active intermediate** A^* is an unstable, high-energy molecule (a free radical, an excited molecule, an ion…). It is destroyed almost as fast as it is formed, so its concentration stays **very small** and **cannot be measured**. On the energy diagram it sits in a shallow **trough** between two barriers.



(a) elementary: one barrier



(b) with an active intermediate

2.2 The Pseudo-Steady-State Hypothesis (PSSH)

PSSH

The **net** rate of formation of every active intermediate is zero:

$$r_{A^*} = \sum_{i=1}^{N} r_{iA^*} = 0.$$

Physical meaning: A^* is formed and consumed at (almost) the same rate, so its tiny concentration changes very little compared with the reactants. PSSH lets you **solve algebraically for** C_{A^*} and remove it from the rate law.

2.3 Relative rates within one step (the most common source of error)

For an elementary step $aA + bB \to cC$: $\dfrac{r_A}{-a} = \dfrac{r_B}{-b} = \dfrac{r_C}{c}$. The rate constant is always defined *with respect to one species*. When you write the PSSH balance for the intermediate, convert every step's rate to the intermediate:

- $\mathrm{Br_2} \xrightarrow{k_1} 2\,\mathrm{Br}\bullet$, with k_1 defined w.r.t. $\mathrm{Br_2}$: $-r_{1,\mathrm{Br_2}} = k_1 C_{\mathrm{Br_2}}$, so $r_{1,\mathrm{Br}\bullet} = +2k_1 C_{\mathrm{Br_2}}$ (factor 2).
- $\mathrm{NO_3}^* + \mathrm{NO} \xrightarrow{k_3} 2\,\mathrm{NO_2}$, with k_3 defined w.r.t. $\mathrm{NO_2}$: $r_{3,\mathrm{NO_2}} = k_3 C_{\mathrm{NO_3^*}} C_{\mathrm{NO}}$, so $r_{3,\mathrm{NO_3^*}} = -\frac{1}{2} k_3 C_{\mathrm{NO_3^*}} C_{\mathrm{NO}}$ (factor $\frac{1}{2}$).

Common mistake

Forgetting the stoichiometric factor (2 or $\frac{1}{2}$) when a step makes or uses *two* of a species. Always write which species k_i refers to before you write the rate.

2.4 Example 1: Lindemann mechanism, $-r_A = \dfrac{kC_A^2}{1 + k'C_A}$

Experiment: the gas-phase decomposition $\mathrm{A} \longrightarrow \mathrm{B} + \mathrm{C}$ follows $-r_A = \dfrac{kC_A^2}{1 + k'C_A}$. **Task:** propose a mechanism.

Reading the rate law: C_A is in the denominator, so A collides with the intermediate. The "1" in the denominator means A^* also decomposes on its own. The C_A^2 in the numerator means A^* is made by an A–A collision.

Mechanism:

$$(1)\ \ A + A \xrightarrow{k_1} A^* + A \qquad (2)\ \ A^* + A \xrightarrow{k_2} A + A \qquad (3)\ \ A^* \xrightarrow{k_3} B + C$$

Step (1) activates A by collision. Step (2) deactivates A^* by collision. Step (3) decomposes A^* into products. All three k's are defined with respect to A^*.

Step 1. Rate laws of each step, written for A^* (each step is elementary, so its rate law follows from its stoichiometry):

$$r_{1A^*} = k_1 C_A^2, \qquad r_{2A^*} = -k_2 C_A C_{A^*}, \qquad r_{3A^*} = -k_3 C_{A^*}.$$

Step 2. Rate of formation of product. B is formed only in step (3), so

$$r_B = r_{3B} = -r_{3A^*} = k_3 C_{A^*}. \tag{a}$$

This contains C_{A^*}, which cannot be measured, so we must eliminate it.

Step 3. PSSH on A^*:

$$r_{A^*} = r_{1A^*} + r_{2A^*} + r_{3A^*} = k_1 C_A^2 - k_2 C_A C_{A^*} - k_3 C_{A^*} = 0.$$

Step 4. Solve for C_{A^*}. Collect the C_{A^*} terms: $k_1 C_A^2 = C_{A^*}(k_3 + k_2 C_A)$, so

$$C_{A^*} = \frac{k_1 C_A^2}{k_3 + k_2 C_A}. \tag{b}$$

Step 5. Substitute (b) into (a), then use A $\longrightarrow$ B + C ($-r_A = r_B$):

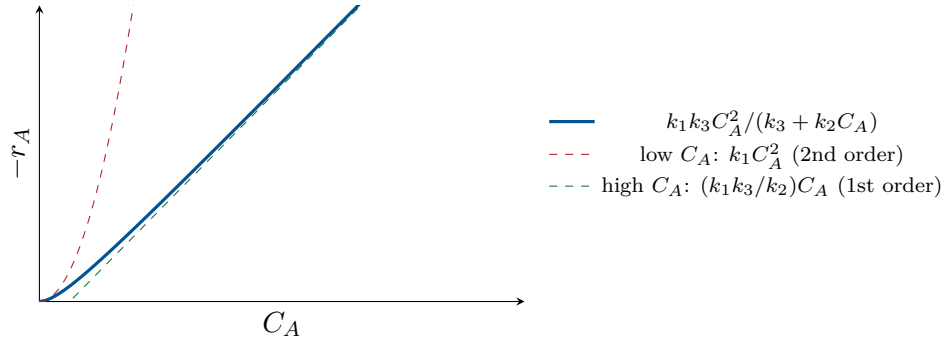
$$-r_A = r_B = \frac{k_1 k_3 C_A^2}{k_3 + k_2 C_A}.$$

Step 6. Match the experimental form. Divide the numerator and denominator by k_3:

$$-r_A = \frac{k_1 C_A^2}{1 + (k_2/k_3) C_A} = \frac{k C_A^2}{1 + k' C_A}, \qquad k = k_1, \quad k' = \frac{k_2}{k_3}. \ \checkmark$$

Limiting cases (a favourite exam question):

- **High** C_A ($k_2 C_A \gg k_3$): $-r_A \approx \dfrac{k_1 k_3}{k_2} C_A$, **apparent first order**. Collisions are so frequent that activation and deactivation are in equilibrium, and step (3) is slow.
- **Low** C_A ($k_3 \gg k_2 C_A$): $-r_A \approx k_1 C_A^2$, **apparent second order**. Every A^* formed decomposes, so the activating collision (1) is rate-limiting.



2.5 Why do so many reactions look elementary? (Collision with an inert)

If the activating partner is an inert I (or the solvent) instead of A:

$$(1)\ A + I \xrightarrow{k_1} A^* + I \qquad (2)\ A^* + I \xrightarrow{k_2} A + I \qquad (3)\ A^* \xrightarrow{k_3} B + C$$

PSSH: $k_1 C_A C_I - k_2 C_I C_{A^*} - k_3 C_{A^*} = 0 \Rightarrow C_{A^*} = \dfrac{k_1 C_A C_I}{k_3 + k_2 C_I}$, so

$$-r_A = k_3 C_{A^*} = \frac{k_1 k_3 C_A C_I}{k_3 + k_2 C_I} = \frac{k C_A C_I}{1 + k' C_I}.$$

When C_I is large and essentially constant ($k_2 C_I \gg k_3$), this becomes $-r_A = \dfrac{k_1 k_3}{k_2} C_A$, i.e. **first order**, which looks exactly like an elementary rate law.

Slide typos

One slide prints the target as $-r_A = kC_A/(1 + k'C_I)$ (missing C_I in the numerator), and another prints the denominator as "$k_2 + k_2 C_I$". The correct result is $\dfrac{k_1 k_3 C_A C_I}{k_3 + k_2 C_I}$.

2.6 Rate that decreases with temperature: $2\,NO + O_2 \longrightarrow 2\,NO_2$

Observation: the rate law is $r_{NO_2} = k_{NO_2} C_{NO}^2 C_{O_2}$ (it looks third-order elementary), but the rate **decreases** as T increases. A true elementary step cannot do that, because its $E > 0$.

Mechanism:

$$(1)\ NO + O_2 \xrightarrow{k_1} NO_3{}^* \qquad (2)\ NO_3{}^* \xrightarrow{k_2} NO + O_2 \qquad (3)\ NO_3{}^* + NO \xrightarrow{k_3} 2\,NO_2$$

Step 1. Rates for each step (k_1, k_2 w.r.t. NO; k_3 w.r.t. NO_2):

$$r_{1,NO} = -k_1 C_{NO} C_{O_2}, \quad r_{2,NO} = k_2 C_{NO_3^*}, \quad r_{3,NO_2} = k_3 C_{NO_3^*} C_{NO}.$$

Step 2. Convert each to NO_3^*: $r_{1,NO_3^*} = -r_{1,NO} = k_1 C_{NO} C_{O_2}$; $r_{2,NO_3^*} = -r_{2,NO} = -k_2 C_{NO_3^*}$; $r_{3,NO_3^*} = -\frac{1}{2} r_{3,NO_2} = -\frac{k_3}{2} C_{NO_3^*} C_{NO}$.

Step 3. PSSH on NO_3^*:

$$k_1 C_{NO} C_{O_2} - k_2 C_{NO_3^*} - \frac{k_3}{2} C_{NO_3^*} C_{NO} = 0 \Rightarrow C_{NO_3^*} = \frac{k_1 C_{NO} C_{O_2}}{k_2 + \frac{k_3}{2} C_{NO}}.$$

Step 4. Approximation. NO_3^* is very unstable, so it falls apart (step 2) much faster than it meets an NO molecule: $k_2 \gg \frac{k_3}{2} C_{NO}$. Then $C_{NO_3^*} \approx \dfrac{k_1}{k_2} C_{NO} C_{O_2}$ (steps 1 and 2 are effectively at equilibrium, $K_1 = k_1/k_2$).

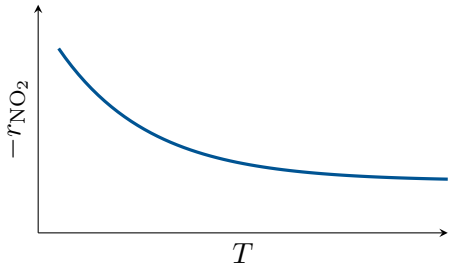
Step 5. Rate of NO_2 formation:

$$r_{NO_2} = k_3 C_{NO_3^*} C_{NO} = \frac{k_1 k_3}{k_2} C_{NO}^2 C_{O_2} \quad \Rightarrow \quad k_{NO_2} = \frac{k_1 k_3}{k_2}.$$

Why the rate falls with T: each $k_i = A_i e^{-E_i/RT}$, so

$$k_{NO_2} = \frac{A_1 A_3}{A_2} \exp\left[-\frac{E_1 + E_3 - E_2}{RT}\right], \qquad E_{app} = E_1 + E_3 - E_2.$$

If the decomposition barrier E_2 is larger than $E_1 + E_3$, then $E_{app} < 0$ and the rate **decreases** with T. Physically, a higher T destroys NO_3^* (step 2) faster than it speeds up the steps that use it.



Apparent activation energy

If $k_{app} = \dfrac{\prod k_{num}}{\prod k_{den}}$, then $E_{app} = \sum E_{num} - \sum E_{den}$. It can be negative.

2.7 Seven steps to find (or check) a mechanism

1. Assume one or more active intermediates.
2. Postulate a mechanism, using the rate law as a guide (denominator species collide with the intermediate).
3. Model each step as an **elementary** reaction.
4. Write the rate law for each step.
5. Apply the PSSH to each intermediate.
6. Eliminate the intermediate concentrations (C_{A^*}, $C_{H\bullet}$, ...).
7. Compare with the experimental data (check limits, orders and the sign of E_{app}).

3 Lecture 7 (Part 1): Chain Reactions, $H_2 + Br_2 \longrightarrow 2\,HBr$

Fogler: 3rd ed. §7.1; HW3 Problem 2

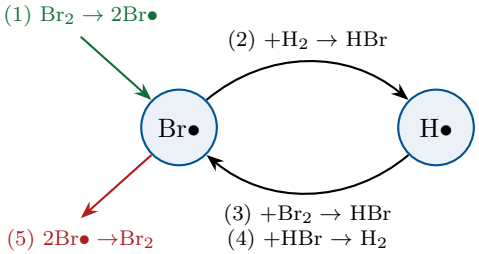
Observed rate law:

$$r_{HBr} = \frac{k' C_{H_2} C_{Br_2}^{3/2}}{C_{Br_2} + k'' C_{HBr}}.$$

Clues: the fractional order $\frac{3}{2}$ suggests Br_2 splits into two radicals (initiation). HBr in the denominator means HBr attacks an intermediate (inhibition by product). The intermediates are H• and Br•.

3.1 Mechanism

Initiation	$Br_2 \xrightarrow{k_1} 2\,Br\bullet$	(1)
Propagation	$Br\bullet + H_2 \xrightarrow{k_2} HBr + H\bullet$	(2)
Propagation	$H\bullet + Br_2 \xrightarrow{k_3} HBr + Br\bullet$	(3)
Inhibition	$H\bullet + HBr \xrightarrow{k_4} H_2 + Br\bullet$	(4)
Termination	$2\,Br\bullet \xrightarrow{k_5} Br_2$	(5)



Propagation is a cycle: each loop makes 2 HBr and gives back Br•.

Here k_1 is w.r.t. Br_2, k_5 is w.r.t. Br• (so $r_{5,Br\bullet} = -k_5 C_{Br\bullet}^2$), and k_2–k_4 are w.r.t. HBr (or the radical, which is the same thing because every coefficient is 1).

3.2 Derivation

Step 1. Net rate of HBr formation. HBr is made in (2) and (3) and consumed in (4):

$$r_{HBr} = k_2 C_{Br\bullet} C_{H_2} + k_3 C_{H\bullet} C_{Br_2} - k_4 C_{H\bullet} C_{HBr}. \tag{i}$$

We need $C_{H\bullet}$ and $C_{Br\bullet}$.

Step 2. PSSH on H• (made in 2, used in 3 and 4):

$$0 = k_2 C_{Br\bullet} C_{H_2} - k_3 C_{H\bullet} C_{Br_2} - k_4 C_{H\bullet} C_{HBr} \;\Rightarrow\; C_{H\bullet} = \frac{k_2 C_{Br\bullet} C_{H_2}}{k_3 C_{Br_2} + k_4 C_{HBr}}. \tag{ii}$$

Step 3. PSSH on Br• (made in 1, 3 and 4; used in 2 and 5):

$$0 = \underbrace{2k_1 C_{Br_2}}_{\text{(1), factor 2}} \underbrace{- k_2 C_{Br\bullet} C_{H_2} + k_3 C_{H\bullet} C_{Br_2} + k_4 C_{H\bullet} C_{HBr}}_{=0 \text{ by the H• balance (ii)}} - k_5 C_{Br\bullet}^2.$$

The middle group is exactly minus the H• balance, so it is zero. That leaves

$$2k_1 C_{Br_2} = k_5 C_{Br\bullet}^2 \;\Rightarrow\; C_{Br\bullet} = \left(\frac{2k_1}{k_5}\right)^{1/2} C_{Br_2}^{1/2}. \tag{iii}$$

In words: *initiation and termination balance each other.* This is where the $\frac{1}{2}$ order comes from.

Step 4. Simplify (i) using (ii). From the H• balance, $k_2 C_{Br\bullet} C_{H_2} - k_4 C_{H\bullet} C_{HBr} = k_3 C_{H\bullet} C_{Br_2}$. Substituting into (i):

$$r_{HBr} = 2k_3 C_{H\bullet} C_{Br_2} = 2k_3 C_{Br_2} \cdot \frac{k_2 C_{Br\bullet} C_{H_2}}{k_3 C_{Br_2} + k_4 C_{HBr}} = \frac{2k_2 C_{Br\bullet} C_{H_2} C_{Br_2}}{C_{Br_2} + (k_4/k_3) C_{HBr}}.$$

(You can also substitute (ii) and (iii) straight into (i) and simplify; the result is the same, with more algebra.)

Step 5. Insert (iii):

$$\boxed{r_{HBr} = \frac{2k_2 \left(\frac{2k_1}{k_5}\right)^{1/2} C_{H_2} C_{Br_2}^{3/2}}{C_{Br_2} + (k_4/k_3) C_{HBr}} = \left(\frac{8k_1 k_2^2}{k_5}\right)^{1/2} \frac{C_{H_2} C_{Br_2}^{3/2}}{C_{Br_2} + (k_4/k_3) C_{HBr}}}$$

so $k' = 2k_2\sqrt{2k_1/k_5}$ and $k'' = k_4/k_3$. This matches the experiment. ✓

3.3 The rate order changes with conditions

Condition	Dominant denominator term	Apparent rate law
Start of reaction (no HBr yet)	C_{Br_2}	$r_{HBr,0} = k' C_{H_2} C_{Br_2}^{1/2}$
High C_{Br_2} ($C_{Br_2} \gg k'' C_{HBr}$)	C_{Br_2}	$\propto C_{H_2} C_{Br_2}^{1/2}$
High C_{HBr} / low C_{Br_2}	$k'' C_{HBr}$	$\propto C_{H_2} C_{Br_2}^{3/2} C_{HBr}^{-1}$

Exam tip

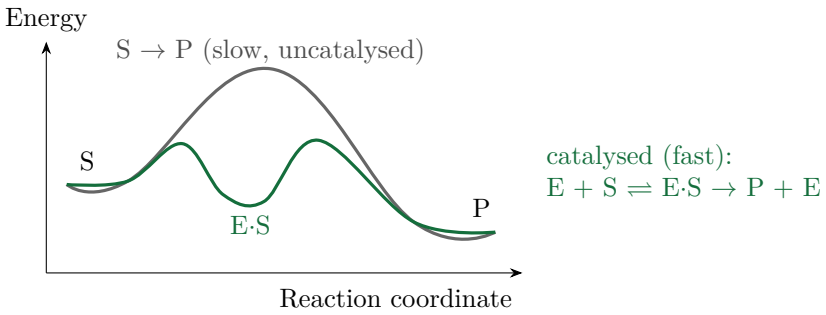
For a chain mechanism, add the PSSH equations for all radicals: the propagation steps cancel, and you are left with **rate of initiation = rate of termination**. That gives the radical concentration almost for free.

4 Lecture 7 (Part 2): Enzyme Kinetics

Fogler: 3rd ed. §7.4–7.5; slides cite pp. 401–409, Ex. 7-3

4.1 Fundamentals

- Enzymes are protein catalysts. They give the substrate S a **faster pathway** to the product P through an enzyme–substrate complex E·S, and the enzyme is regenerated.
- They are highly **specific**: a given enzyme catalyses essentially one reaction. Example: urease decomposes urea only, $NH_2CONH_2 + \text{urease} \xrightarrow{H_2O} 2\,NH_3 + CO_2 + \text{urease}$.



4.2 Michaelis–Menten derivation (urease mechanism)

Mechanism (W = water, always in large excess):

$$(1)\ E + S \xrightarrow{k_1} E\cdot S \qquad (2)\ E\cdot S \xrightarrow{k_2} E + S \qquad (3)\ E\cdot S + W \xrightarrow{k_3} P + E$$

Enzyme balance (the enzyme is either free or complexed): $C_{E,t} = C_E + C_{E\cdot S}$.

Step 1. Rate of product formation: $r_P = k_3 C_{E\cdot S} C_W$. $C_{E\cdot S}$ is not measurable.

Step 2. PSSH on E·S:

$$r_{E\cdot S} = k_1 C_E C_S - k_2 C_{E\cdot S} - k_3 C_{E\cdot S} C_W = 0.$$

Step 3. Eliminate the free enzyme using $C_E = C_{E,t} - C_{E\cdot S}$:

$$k_1(C_{E,t} - C_{E\cdot S})C_S = C_{E\cdot S}(k_2 + k_3 C_W) \;\Rightarrow\; C_{E\cdot S} = \frac{k_1 C_{E,t} C_S}{k_1 C_S + k_2 + k_3 C_W}.$$

Step 4. Substitute:

$$r_P = \frac{k_1 k_3 C_{E,t} C_W C_S}{k_1 C_S + k_2 + k_3 C_W}.$$

Step 5. Group constants. Divide the top and bottom by k_1, and define

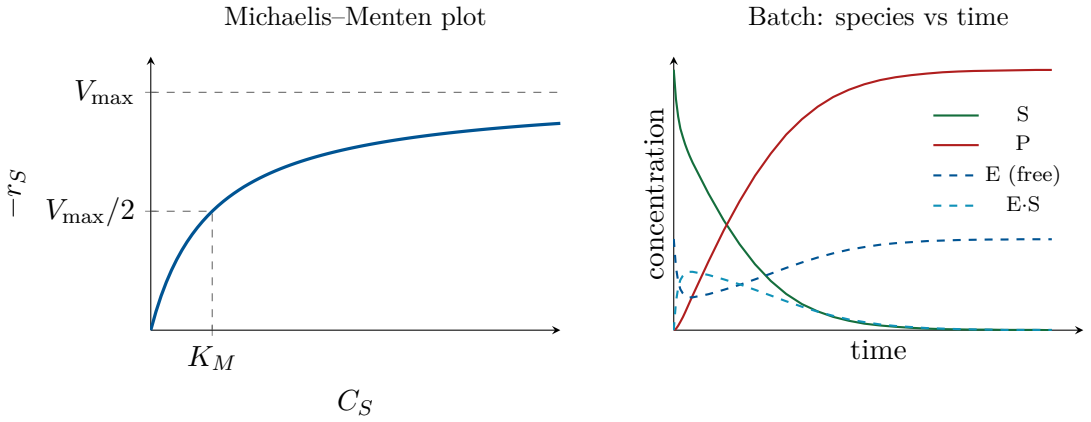
$$k_{\text{cat}} = k_3' = k_3 C_W \text{ (turnover number)}, \qquad K_M = \frac{k_2 + k_3'}{k_1} \text{ (Michaelis constant)}, \qquad V_{\max} = k_{\text{cat}} C_{E,t}.$$

Michaelis–Menten equation

$$-r_S = r_P = \frac{V_{\max} C_S}{K_M + C_S} = \frac{k_{\text{cat}} C_{E,t} C_S}{K_M + C_S}$$

- $C_S \ll K_M$: $-r_S \approx \dfrac{V_{\max}}{K_M} C_S$, **first order**.
- $C_S \gg K_M$: $-r_S \approx V_{\max}$, **zero order** (the enzyme is saturated).
- At $C_S = K_M$: $-r_S = V_{\max}/2$. K_M **is the substrate concentration at half the maximum rate** (set $\frac{V_{\max}}{2} = \frac{V_{\max} S_{1/2}}{K_M + S_{1/2}}$, which gives $K_M = S_{1/2}$). A small K_M means a high affinity of the enzyme for S.

Turnover number k_{cat}: the number of substrate molecules converted per enzyme molecule per unit time (molecules/molecule/s). Example: catalase, $H_2O_2 + E \longrightarrow H_2O + O + E$, has $k_{\text{cat}} = 4 \times 10^7$ s^{-1}.



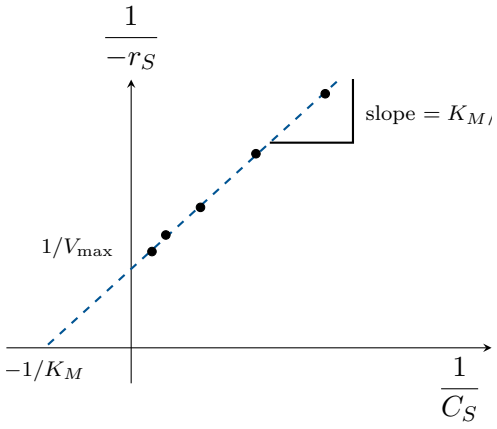
Right: $C_E + C_{E\cdot S} = C_{E,t}$ at all times. After a very short start-up, $C_{E\cdot S}$ changes slowly compared with S and P, which is why the PSSH works.

4.3 Finding $V_{\max}$ and K_M from data: linearised plots

Lineweaver–Burk (double reciprocal). Invert the M–M equation:

$$\frac{1}{-r_S} = \frac{K_M + C_S}{V_{\max} C_S} = \frac{1}{V_{\max}} + \frac{K_M}{V_{\max}} \cdot \frac{1}{C_S}.$$

Plot $1/(-r_S)$ against $1/C_S$: a straight line with **intercept** $1/V_{\max}$, **slope** $K_M/V_{\max}$, and x**-intercept** $-1/K_M$. Then $V_{\max} = 1/\text{intercept}$ and $K_M = \text{slope} \times V_{\max}$.



Two other linear forms (Fogler) are less sensitive to errors in the low-C_S points, which dominate a Lineweaver–Burk plot:

- **Hanes–Woolf:** $\dfrac{C_S}{-r_S} = \dfrac{K_M}{V_{\max}} + \dfrac{1}{V_{\max}} C_S$ (plot $C_S/(-r_S)$ vs C_S).
- **Eadie–Hofstee:** $-r_S = V_{\max} - K_M \dfrac{-r_S}{C_S}$ (plot $-r_S$ vs $-r_S/C_S$; slope $= -K_M$).

4.4 Enzyme reactions in a batch reactor (very exam-friendly)

Mole balance, constant volume: $\dfrac{dC_S}{dt} = r_S = -\dfrac{V_{\max} C_S}{K_M + C_S}$.

Step 1. Separate variables: $-\dfrac{K_M + C_S}{C_S}\, dC_S = V_{\max}\, dt$.

Step 2. Integrate from C_{S0} at $t = 0$ to C_S at t: $K_M \ln \dfrac{C_{S0}}{C_S} + (C_{S0} - C_S) = V_{\max}\, t$.

Step 3. With $C_S = C_{S0}(1 - X)$:

Batch enzymatic reactor

$$t = \frac{K_M}{V_{\max}} \ln \frac{1}{1 - X} + \frac{C_{S0} X}{V_{\max}}$$

The first term is the first-order (low-C_S) part and the second is the zero-order (saturated) part. Rearranged for data analysis: $\dfrac{1}{t} \ln \dfrac{C_{S0}}{C_S} = \dfrac{V_{\max}}{K_M} - \dfrac{C_{S0} - C_S}{K_M\, t}$.

CSTR: $V = \dfrac{v_0(C_{S0} - C_S)}{-r_S} = \dfrac{v_0(C_{S0} - C_S)(K_M + C_S)}{V_{\max} C_S}$.

4.5 Enzyme inhibition

An inhibitor I binds the enzyme and makes it inactive. There are three types:

Type	Inhibitor binds to	Inactive complex(es)
1. Competitive	free enzyme only (competes with S for the active site)	$\mathrm{E}+\mathrm{I} \rightleftharpoons \mathrm{I}\cdot\mathrm{E}$
2. Uncompetitive	the E·S complex only	$\mathrm{E}\cdot\mathrm{S}+\mathrm{I} \rightleftharpoons \mathrm{I}\cdot\mathrm{E}\cdot\mathrm{S}$
3. Noncompetitive	both E and E·S (binds a different site)	$\mathrm{E}+\mathrm{I} \rightleftharpoons \mathrm{I}\cdot\mathrm{E}$
		$\mathrm{E}\cdot\mathrm{S}+\mathrm{I} \rightleftharpoons \mathrm{I}\cdot\mathrm{E}\cdot\mathrm{S}$
		$\mathrm{I}\cdot\mathrm{E}+\mathrm{S} \rightleftharpoons \mathrm{I}\cdot\mathrm{E}\cdot\mathrm{S}$

In all three, use the same recipe: (1) PSSH or equilibrium for each complex; (2) express every complex in terms of C_E; (3) total enzyme balance to find C_E; (4) $-r_S = r_P = k_3 C_{E\cdot S}$; (5) invert for the Lineweaver–Burk form. Throughout, $K_M = \dfrac{k_2+k_3}{k_1}$ (water folded into k_3) and $V_{\max} = k_3 C_{E,t}$.

1. Competitive inhibition (derived in lecture)

$$E + S \underset{k_2}{\overset{k_1}{\rightleftharpoons}} E\cdot S \xrightarrow{k_3} E + P, \qquad E + I \underset{k_5}{\overset{k_4}{\rightleftharpoons}} E\cdot I \text{ (inactive)}$$

Step 1. PSSH on E·S: $0 = k_1 C_S C_E - (k_2+k_3) C_{E\cdot S} \Rightarrow C_{E\cdot S} = \dfrac{C_S C_E}{K_M}$.

Step 2. PSSH on E·I: $0 = k_4 C_I C_E - k_5 C_{E\cdot I} \Rightarrow C_{E\cdot I} = \dfrac{C_I C_E}{K_I}$, with $K_I = \dfrac{k_5}{k_4}$.

Step 3. Total enzyme: $C_{E,t} = C_E + C_{E\cdot S} + C_{E\cdot I} = C_E\left(1 + \dfrac{C_S}{K_M} + \dfrac{C_I}{K_I}\right)$.

Step 4. Rate:

$$-r_S = k_3 C_{E\cdot S} = \frac{k_3 C_S C_E}{K_M} = \frac{k_3 C_{E,t} C_S / K_M}{1 + \frac{C_S}{K_M} + \frac{C_I}{K_I}} = \frac{V_{\max} C_S}{C_S + K_M\left(1 + \frac{C_I}{K_I}\right)}.$$

Step 5. Invert: $\dfrac{1}{-r_S} = \dfrac{1}{V_{\max}} + \dfrac{K_M}{V_{\max}}\left(1 + \dfrac{C_I}{K_I}\right)\dfrac{1}{C_S}$.

Meaning: the intercept ($1/V_{\max}$) is unchanged and the slope increases. The apparent K_M becomes $K_M(1 + C_I/K_I)$. At a very high C_S the substrate out-competes the inhibitor, so $V_{\max}$ can still be reached.

2. Uncompetitive inhibition

$$E + S \rightleftharpoons E\cdot S \xrightarrow{k_3} E + P, \qquad E\cdot S + I \underset{k_5}{\overset{k_4}{\rightleftharpoons}} I\cdot E\cdot S \text{ (inactive)}$$

Step 1. PSSH on E·S. The I-binding step is at equilibrium, so its net rate is zero and it drops out of the E·S balance: $C_{E\cdot S} = \dfrac{C_S C_E}{K_M}$.

Step 2. PSSH on I·E·S: $0 = k_4 C_I C_{E\cdot S} - k_5 C_{I\cdot E\cdot S} \Rightarrow C_{I\cdot E\cdot S} = \dfrac{C_I C_{E\cdot S}}{K_I} = \dfrac{C_I C_S C_E}{K_I K_M}$.

Step 3. Total enzyme:

$$C_{E,t} = C_E + C_{E\cdot S} + C_{I\cdot E\cdot S} = C_E\left[1 + \frac{C_S}{K_M}\left(1 + \frac{C_I}{K_I}\right)\right].$$

Step 4. Rate (multiply the top and bottom by K_M):

$$-r_S = \frac{k_3 C_S C_E}{K_M} = \frac{V_{\max} C_S}{K_M + C_S\left(1 + \frac{C_I}{K_I}\right)}.$$

Step 5. Invert:

$$\frac{1}{-r_S} = \frac{K_M + C_S(1 + C_I/K_I)}{V_{\max} C_S} = \frac{1}{V_{\max}}\left(1 + \frac{C_I}{K_I}\right) + \frac{K_M}{V_{\max}} \cdot \frac{1}{C_S}.$$

Meaning: the slope is unchanged and the intercept increases, so the lines are **parallel**. Both the apparent $V_{\max}$ and the apparent K_M are divided by $(1 + C_I/K_I)$.

3. Noncompetitive inhibition (HW3 Problem 3)

I binds to a site different from the active site, so it can attach to free E or to E·S, and S can still bind to I·E:

$$E + S \rightleftharpoons E \cdot S \xrightarrow{k_3} E + P, \qquad E + I \rightleftharpoons I \cdot E \ (K_I),$$
$$E \cdot S + I \rightleftharpoons I \cdot E \cdot S \ (K_I), \qquad I \cdot E + S \rightleftharpoons I \cdot E \cdot S \ \text{(inactive)}.$$

Assume I binds equally well to E and E·S (the same K_I), and that S binds I·E as it binds E (the same K_M).

Step 1. Express every complex through C_E:

$$C_{E\cdot S} = \frac{C_S C_E}{K_M}, \qquad C_{I\cdot E} = \frac{C_I C_E}{K_I}, \qquad C_{I\cdot E\cdot S} = \frac{C_I C_{E\cdot S}}{K_I} = \frac{C_I C_S C_E}{K_I K_M}.$$

Step 2. Total enzyme. The four terms factor neatly:

$$C_{E,t} = C_E\left[1 + \frac{C_S}{K_M} + \frac{C_I}{K_I} + \frac{C_I C_S}{K_I K_M}\right] = C_E\left(1 + \frac{C_S}{K_M}\right)\left(1 + \frac{C_I}{K_I}\right).$$

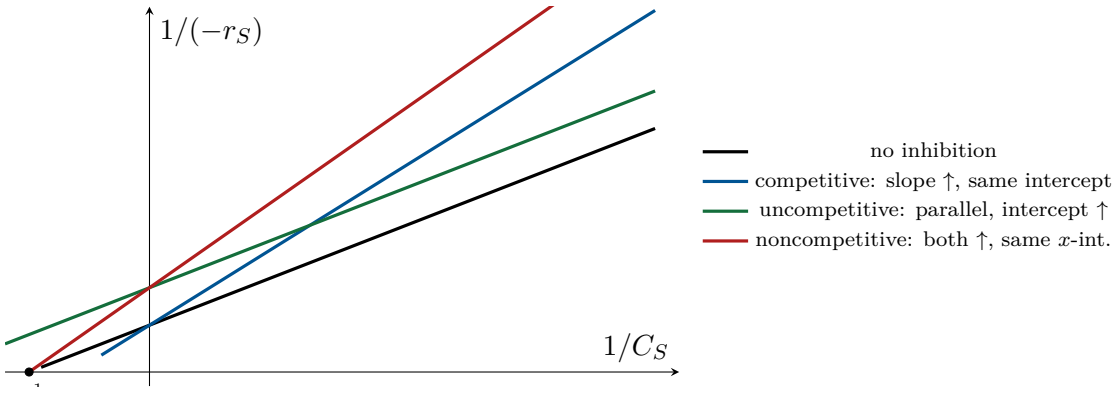
Step 3. Rate:

$$-r_S = \frac{k_3 C_S C_E}{K_M} = \frac{k_3 C_{E,t} C_S}{K_M\left(1 + \frac{C_S}{K_M}\right)\left(1 + \frac{C_I}{K_I}\right)} = \frac{V_{\max} C_S}{(C_S + K_M)\left(1 + \frac{C_I}{K_I}\right)}.$$

Step 4. Invert:

$$\frac{1}{-r_S} = \frac{1}{V_{\max}}\left(1 + \frac{C_I}{K_I}\right) + \frac{K_M}{V_{\max}}\left(1 + \frac{C_I}{K_I}\right)\frac{1}{C_S}.$$

Meaning: the slope and the intercept both increase by the same factor $(1 + C_I/K_I)$. The x-intercept $(-1/K_M)$ is unchanged, so the lines meet on the x-axis. The apparent $V_{\max}$ drops and K_M stays the same.



Inhibition summary: write $\phi = 1 + C_I/K_I$

Type	$-r_S$	L–B intercept	L–B slope	apparent $V_{\max}$, K_M
None	$\dfrac{V_{\max}C_S}{K_M + C_S}$	$\dfrac{1}{V_{\max}}$	$\dfrac{K_M}{V_{\max}}$	$V_{\max}$, K_M
Competitive	$\dfrac{V_{\max}C_S}{C_S + K_M\phi}$	same	$\times\phi$	$V_{\max}$, $K_M\phi$
Uncompetitive	$\dfrac{V_{\max}C_S}{K_M + C_S\phi}$	$\times\phi$	same	$V_{\max}/\phi$, K_M/ϕ
Noncompetitive	$\dfrac{V_{\max}C_S}{(C_S + K_M)\phi}$	$\times\phi$	$\times\phi$	$V_{\max}/\phi$, K_M

Exam tip

To identify the inhibition type from data: compute $1/(-r_S)$ and $1/C_S$ with and without inhibitor, fit both lines, and compare the slopes and intercepts. The *ratio* (inhibited/uninhibited) of whichever changed equals $\phi = 1 + C_I/K_I$, so $K_I = C_I/(\phi - 1)$.

5 Lectures 8–9: Catalysis and Langmuir–Hinshelwood Kinetics

Fogler: 4th ed. Ch. 10, pp. 645–700; 5th ed. Ch. 10 (starts p. 399)

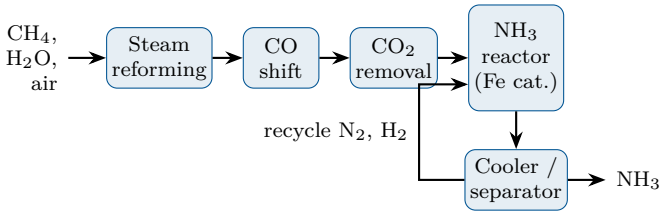
5.1 Catalysts and an industrial example

- A **catalyst** changes the rate of a reaction (by giving a lower-energy pathway) without being consumed. It does **not** change the equilibrium.
- **Heterogeneous** catalysis: the catalyst is a solid and the reactants are gases or liquids. The reaction happens on the **surface**, mostly inside the **pores** of a porous pellet, which is where nearly all of the area is.
- A catalyst is often a small amount of **active** material (Pt, Ni, Fe…) dispersed on a high-area **support** (Al_2O_3, SiO_2, zeolites, carbon).

Haber–Bosch ammonia synthesis ($N_2 + 3\,H_2 \longrightarrow 2\,NH_3$, promoted Fe catalyst with K_2O/Al_2O_3):

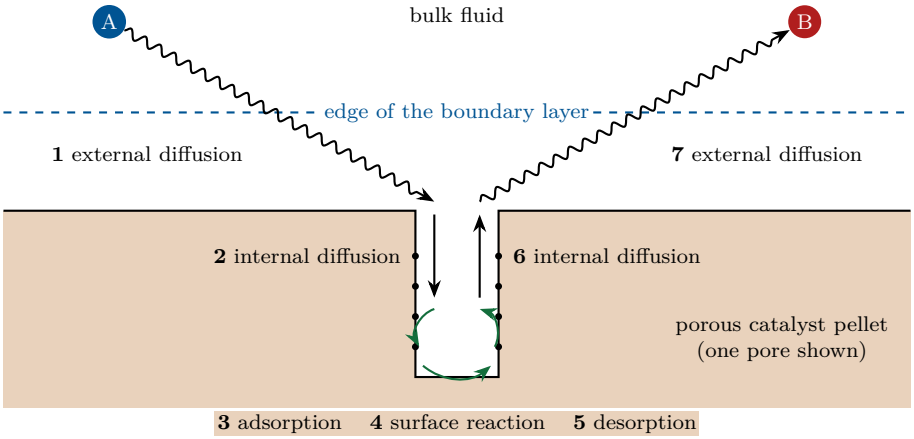
(1) $N_2(g) \longrightarrow N_2(ads)$
(2) $N_2(ads) \longrightarrow 2\,N(ads)$
(3) $H_2(g) \longrightarrow H_2(ads)$
(4) $H_2(ads) \longrightarrow 2\,H(ads)$
(5) $N(ads) + 3\,H(ads) \longrightarrow NH_3(ads)$
(6) $NH_3(ads) \longrightarrow NH_3(g)$

Adsorb, then dissociate, then react on the surface, then desorb.



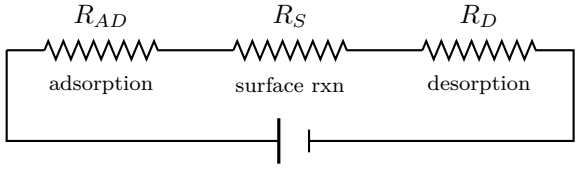
5.2 The seven steps of a heterogeneous catalytic reaction

Fogler: 4th ed. p. 656



1. **External diffusion** of A from the bulk, through the boundary layer, to the pellet surface.
2. **Internal diffusion** of A through the pores to the catalytic surface.
3. **Adsorption** of A onto a site: $\mathrm{A} + \mathrm{S} \rightleftharpoons \mathrm{A}\cdot\mathrm{S}$.
4. **Surface reaction**: $\mathrm{A}\cdot\mathrm{S} \rightleftharpoons \mathrm{B}\cdot\mathrm{S}$.
5. **Desorption** of B: $\mathrm{B}\cdot\mathrm{S} \rightleftharpoons \mathrm{B} + \mathrm{S}$.
6. **Internal diffusion** of B to the pore mouth.
7. **External diffusion** of B to the bulk.

At steady state all the steps run in series at the same rate. The **slowest** step controls the overall rate, like the largest resistor in a series circuit. In Lectures 8–9 we assume diffusion (1, 2, 6, 7) is fast and focus on steps 3–5 (*reaction-limited*). Diffusion limitations come later in the course.



Electrical analogy: the current (rate) is the same through every resistor and is set by the largest one.

5.3 Notation

S	an active site; A·S is A adsorbed on a site
C_t	total concentration of sites (mol sites / g cat), **constant**
C_v	concentration of vacant sites (mol / g cat)
C_{AS}	concentration of sites occupied by A (mol / g cat)
P_A	partial pressure of A in the gas (atm). For liquids use C_A instead.
$K_A = k_A/k_{-A}$	adsorption equilibrium constant of A (atm^{-1})
$-r'_A$	rate per mass of catalyst (mol / g cat / s)

5.4 Adsorption isotherms

Fogler: 4th ed. pp. 661–666

(a) Molecular (non-dissociative) adsorption: $\mathbf{A + S \rightleftharpoons A \cdot S}$

Treat it as an elementary reaction. The rate of collisions with the surface is proportional to P_A, so

$$r_{AD} = \underbrace{k_A P_A C_v}_{\text{adsorption}} - \underbrace{k_{-A} C_{\text{AS}}}_{\text{desorption}} = k_A \left(P_A C_v - \frac{C_{\text{AS}}}{K_A} \right).$$

Step 1. At equilibrium $r_{AD} = 0$, so $C_{\text{AS}} = K_A P_A C_v$.

Step 2. Site balance: $C_t = C_v + C_{\text{AS}} = C_v(1 + K_A P_A) \;\Rightarrow\; C_v = \dfrac{C_t}{1 + K_A P_A}$.

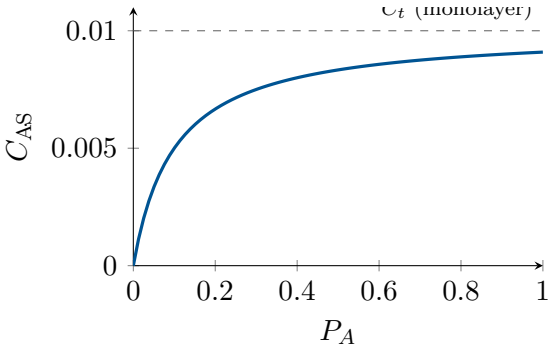
Step 3. Substitute:

Langmuir isotherm (single site)

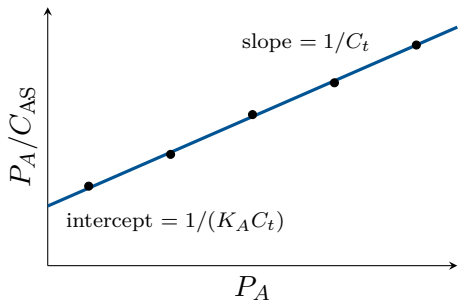
$$C_{\text{AS}} = \frac{K_A P_A C_t}{1 + K_A P_A} \qquad \Longleftrightarrow \qquad \frac{P_A}{C_{\text{AS}}} = \frac{1}{K_A C_t} + \frac{P_A}{C_t}$$

Assumptions: monolayer coverage, a uniform surface (all sites equal), and interaction only between the gas and a site (adsorbed molecules do not interact with each other).

Langmuir isotherm ($C_t = 0.01$, $K_A = 10$)



Linearised: tests for single-single-site Langmuir



(b) Dissociative (atomic) adsorption: $\mathbf{A_2 + 2\,S \rightleftharpoons 2\,A \cdot S}$

Two vacant sites are needed, and two adsorbed atoms must recombine to desorb:

$$r_{AD} = k_A P_{A_2} C_v^2 - k_{-A} C_{\text{AS}}^2 = k_A \left(P_{A_2} C_v^2 - \frac{C_{\text{AS}}^2}{K_A} \right).$$

Step 1. At equilibrium: $C_{\text{AS}}^2 = K_A P_{A_2} C_v^2 \Rightarrow C_{\text{AS}} = (K_A P_{A_2})^{1/2} C_v$.

Step 2. Site balance: $C_t = C_v \left[1 + (K_A P_{A_2})^{1/2} \right]$.

Step 3.

Dissociative Langmuir isotherm

$$C_{\text{AS}} = \frac{(K_A P_{A_2})^{1/2} C_t}{1 + (K_A P_{A_2})^{1/2}} \qquad \Longleftrightarrow \qquad \frac{1}{C_{\text{AS}}} = \frac{1}{C_t} + \frac{1}{C_t \sqrt{K_A}} \cdot \frac{1}{\sqrt{P_{A_2}}}$$

Signature: square roots of pressure. To test, plot $1/C_{\text{AS}}$ against $1/\sqrt{P}$: a straight line means

dissociative.

(c) Several species competing for the same sites

For $\mathrm{A} + \mathrm{S} \rightleftharpoons \mathrm{A}\cdot\mathrm{S}$ and $\mathrm{B} + \mathrm{S} \rightleftharpoons \mathrm{B}\cdot\mathrm{S}$, each one at equilibrium gives $C_{i\mathrm{S}} = K_i P_i C_v$. The site balance $C_t = C_v + C_{\mathrm{AS}} + C_{\mathrm{BS}}$ gives

$$C_v = \frac{C_t}{1 + K_A P_A + K_B P_B}, \qquad C_{\mathrm{AS}} = \frac{K_A P_A C_t}{1 + K_A P_A + K_B P_B}.$$

General rule: the site balance always gives $C_v = C_t / \left(1 + \sum_i (\text{adsorbed term of } i)\right)$, where the term is $K_i P_i$ for molecular adsorption and $\sqrt{K_i P_i}$ for dissociative adsorption.

5.5 Surface reactions and desorption

Fogler: 4th ed. pp. 666–669

Type	Step	Rate law (K_S = surface equilibrium constant)
Single site (L–H)	$\mathrm{A}\cdot\mathrm{S} \rightleftharpoons \mathrm{B}\cdot\mathrm{S}$	$r_S = k_S \left(C_{\mathrm{AS}} - \dfrac{C_{\mathrm{BS}}}{K_S} \right)$
Dual site, adjacent vacant site	$\mathrm{A}\cdot\mathrm{S} + \mathrm{S} \rightleftharpoons \mathrm{B}\cdot\mathrm{S} + \mathrm{S}$	$r_S = k_S \left(C_{\mathrm{AS}} C_v - \dfrac{C_{\mathrm{BS}} C_v}{K_S} \right)$
Dual site, two adsorbed species	$\mathrm{A}\cdot\mathrm{S} + \mathrm{B}\cdot\mathrm{S} \rightleftharpoons \mathrm{C}\cdot\mathrm{S} + \mathrm{D}\cdot\mathrm{S}$	$r_S = k_S \left(C_{\mathrm{AS}} C_{\mathrm{BS}} - \dfrac{C_{\mathrm{CS}} C_{\mathrm{DS}}}{K_S} \right)$
Eley–Rideal (gas attacks)	$\mathrm{A}\cdot\mathrm{S} + \mathrm{B(g)} \rightleftharpoons \mathrm{C}\cdot\mathrm{S}$	$r_S = k_S \left(C_{\mathrm{AS}} P_B - \dfrac{C_{\mathrm{CS}}}{K_S} \right)$
Desorption	$\mathrm{C}\cdot\mathrm{S} \rightleftharpoons \mathrm{C} + \mathrm{S}$	$r_D = k_D \left(C_{\mathrm{CS}} - K_C P_C C_v \right)$

Examples: single site, isomerisation; dual site, dehydration of alcohols; two adsorbed species, $\mathrm{CO} + \mathrm{O} \longrightarrow \mathrm{CO_2}$; the variant on two different site types is $\mathrm{A}\cdot\mathrm{S} + \mathrm{B}\cdot\mathrm{S}' \rightleftharpoons \mathrm{C}\cdot\mathrm{S}' + \mathrm{D}\cdot\mathrm{S}$. For desorption, K_C is the *adsorption* constant of C, so a desorption step at equilibrium gives $C_{\mathrm{CS}} = K_C P_C C_v$, the same as adsorption.

5.6 The rate-limiting-step method

Fogler: 4th ed. p. 669; 5th ed. Ch. 10

At steady state the steps are in series, so $-r'_A = r_{AD} = r_S = r_D$. One step is slow (it has a small k and controls the rate). The **other steps are fast**: their k is huge but their rate is finite, so their bracket must be almost zero. They are **effectively at equilibrium**:

$$\frac{r_{\text{fast}}}{k_{\text{fast}}} \approx 0.$$

Langmuir–Hinshelwood protocol (memorise these 7 steps)

1. Propose a mechanism (molecular or dissociative adsorption; single site, dual site, or Eley–Rideal).
2. Write the net rate law for every step.
3. Pick a rate-limiting step (RLS). Its rate law **is** the overall rate.

4. Set every other step at equilibrium ($r/k \approx 0$) and solve for each $C_{i\cdot S}$ in terms of the P_i and C_v.
5. Substitute these into the RLS rate law, so that only C_v and the P_i remain.
6. Write the total site balance and solve for C_v.
7. Substitute C_v, group the constants, and check: at equilibrium, the rate must be zero when $P_B/P_A = K_P$.

Consistency check on the overall equilibrium constant

The overall K_P is the product of the step equilibrium constants, with desorption contributing $1/K_i$. For A $\rightleftharpoons$ B via adsorb–react–desorb: $K_P = \dfrac{K_A K_S}{K_B}$. Your final rate law must contain $(P_A - P_B/K_P)$.

5.7 Example 1: Surface reaction limiting (the most common case, $> 75\%$ of reactions)

Mechanism, A $\rightleftharpoons$ B: $\mathrm{A} + \mathrm{S} \rightleftharpoons \mathrm{A}\cdot\mathrm{S}$ (fast), $\underline{\mathrm{A}\cdot\mathrm{S} \rightleftharpoons \mathrm{B}\cdot\mathrm{S}}$ (RLS), $\mathrm{B}\cdot\mathrm{S} \rightleftharpoons \mathrm{B} + \mathrm{S}$ (fast).

Step 1. **RLS rate law:** $-r'_A = r_S = k_S\left(C_{\mathrm{AS}} - \dfrac{C_{\mathrm{BS}}}{K_S}\right)$.

Step 2. **Fast steps at equilibrium:**

$$\frac{r_{AD}}{k_A} = P_A C_v - \frac{C_{\mathrm{AS}}}{K_A} \approx 0 \Rightarrow C_{\mathrm{AS}} = K_A P_A C_v; \qquad \frac{r_D}{k_D} = C_{\mathrm{BS}} - K_B P_B C_v \approx 0 \Rightarrow C_{\mathrm{BS}} = K_B P_B C_v.$$

Step 3. **Substitute into the RLS:**

$$r_S = k_S C_v\left(K_A P_A - \frac{K_B}{K_S} P_B\right) = k_S K_A C_v\left(P_A - \frac{K_B}{K_A K_S} P_B\right) = k_S K_A C_v\left(P_A - \frac{P_B}{K_P}\right), \quad K_P = \frac{K_A K_S}{K_B}.$$

Step 4. **Site balance:** $C_t = C_v + K_A P_A C_v + K_B P_B C_v \Rightarrow C_v = \dfrac{C_t}{1 + K_A P_A + K_B P_B}$.

Step 5. **Final:**

Surface reaction limiting, single site

$$-r'_A = \frac{k\,(P_A - P_B/K_P)}{1 + K_A P_A + K_B P_B}, \qquad k = k_S K_A C_t, \quad K_P = \frac{K_A K_S}{K_B}.$$

Initial rate (no B in the feed): $-r'_{A0} = \dfrac{k P_{A0}}{1 + K_A P_{A0}}$. It is linear at low P_{A0} and levels off at k/K_A at high P_{A0}.

Slide notation

The slides write $(P_A - K_e P_B)$ with $K_e = K_B/(K_A K_S)$ in Example 1, but $(P_A - P_B/K_e)$ with $K_e = K_A K_S/K_B$ in Example 2. They are reciprocals of each other. These notes use one constant throughout: $K_P = K_A K_S/K_B$, the true equilibrium constant $(P_B/P_A)_{\mathrm{eq}}$, as in Fogler.

5.8 Example 2: Adsorption of A limiting

Same mechanism, but now $\underline{\mathrm{A} + \mathrm{S} \rightleftharpoons \mathrm{A}\cdot\mathrm{S}}$ is the RLS.

Step 1. **RLS rate law:** $-r'_A = r_{AD} = k_A\left(P_A C_v - \dfrac{C_{\mathrm{AS}}}{K_A}\right)$.

Step 2. **Fast steps at equilibrium:**

$$\frac{r_S}{k_S} = C_{\mathrm{AS}} - \frac{C_{\mathrm{BS}}}{K_S} \approx 0 \Rightarrow C_{\mathrm{AS}} = \frac{C_{\mathrm{BS}}}{K_S}; \qquad C_{\mathrm{BS}} = K_B P_B C_v \;\Rightarrow\; C_{\mathrm{AS}} = \frac{K_B P_B}{K_S} C_v.$$

Note that C_{AS} is set by the *products*, not by P_A: A on the surface is in equilibrium with B.

Step 3. **Substitute:** $r_{AD} = k_A C_v\left(P_A - \dfrac{K_B}{K_S K_A} P_B\right) = k_A C_v\left(P_A - \dfrac{P_B}{K_P}\right)$.

Step 4. **Site balance:**

$$C_t = C_v + \frac{K_B P_B}{K_S} C_v + K_B P_B C_v = C_v\left[1 + \frac{K_A}{K_P} P_B + K_B P_B\right] \quad \left(\text{using } \frac{K_B}{K_S} = \frac{K_A}{K_P}\right).$$

Step 5. **Final:**

Adsorption of A limiting

$$-r'_A = \frac{k_A C_t\,(P_A - P_B/K_P)}{1 + \dfrac{K_A}{K_P} P_B + K_B P_B}.$$

Initial rate ($P_B = 0$): $-r'_{A0} = k_A C_t P_{A0}$, **linear in P_{A0} with no levelling off**. There is no $K_A P_A$ term in the denominator.

Slide typo

The slide's denominator, $1 + P_B(K_B + K_e K_B)$ (and $1 + (K_B + K_e K_A)P_B$ in the C_v line), is wrong. The correct denominator is $1 + K_B P_B\left(1 + \frac{1}{K_S}\right) = 1 + \frac{K_A}{K_P} P_B + K_B P_B$.

5.9 Example 3 (for completeness): Desorption of B limiting

Here $\mathrm{A} + \mathrm{S} \rightleftharpoons \mathrm{A}\cdot\mathrm{S}$ and $\mathrm{A}\cdot\mathrm{S} \rightleftharpoons \mathrm{B}\cdot\mathrm{S}$ are fast, so $C_{\mathrm{AS}} = K_A P_A C_v$ and $C_{\mathrm{BS}} = K_S C_{\mathrm{AS}} = K_S K_A P_A C_v$. The RLS is

$$r_D = k_D\,(C_{\mathrm{BS}} - K_B P_B C_v) = k_D K_B C_v\left(\frac{K_S K_A}{K_B} P_A - P_B\right) = k_D K_B K_P C_v\left(P_A - \frac{P_B}{K_P}\right).$$

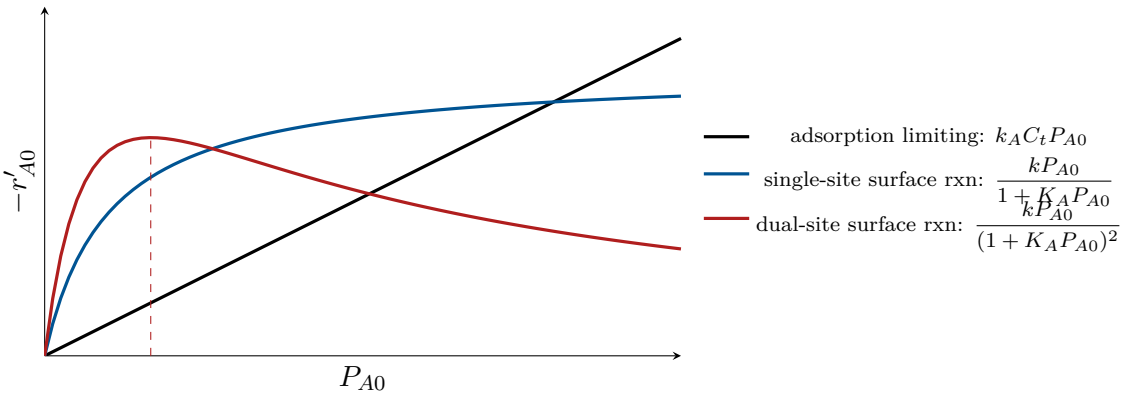
Site balance: $C_t = C_v(1 + K_A P_A + K_S K_A P_A)$, so

$$-r'_A = \frac{k_D K_B K_P C_t\,(P_A - P_B/K_P)}{1 + K_A(1 + K_S) P_A} \quad \Rightarrow \quad -r'_{A0} = \frac{k' P_{A0}}{1 + K_A(1 + K_S) P_{A0}}.$$

The initial rate also levels off at high P_{A0}, and no P_B term appears in the denominator.

5.10 Diagnosing the RLS from initial-rate data

Measure $-r'_{A0}$ against the feed pressure P_{A0} with no products present. The shape of the curve identifies the mechanism:



RLS	$-r'_{A0}$	Linear plot for parameters
Adsorption of A	$k_A C_t P_{A0}$	$-r'_{A0}$ vs P_{A0} (straight through the origin)
Single-site surface reaction	$\dfrac{kP_{A0}}{1+K_AP_{A0}}$	$\dfrac{P_{A0}}{-r'_{A0}} = \dfrac{1}{k} + \dfrac{K_A}{k}P_{A0}$
Dual-site surface reaction	$\dfrac{kP_{A0}}{(1+K_AP_{A0})^2}$	$\sqrt{\dfrac{P_{A0}}{-r'_{A0}}} = \dfrac{1}{\sqrt{k}} + \dfrac{K_A}{\sqrt{k}}P_{A0}$

Dual site, maximum: $\dfrac{d}{dP}\left[\dfrac{kP}{(1+K_AP)^2}\right] = k\dfrac{(1+K_AP)-2K_AP}{(1+K_AP)^3} = 0 \Rightarrow P^* = \dfrac{1}{K_A}$ and $-r'_{\max} = \dfrac{k}{4K_A}$.
Past the maximum, A covers so many sites that there are no *vacant* neighbouring sites left for the reaction.

5.11 Bimolecular L–H mechanism (Lecture 9): A + W $\rightleftharpoons$ B

(The slides use liquid-phase concentrations C_i instead of P_i. The algebra is identical.)

$$\mathrm{A} + \mathrm{S} \rightleftharpoons \mathrm{A}\cdot\mathrm{S}\ (K_A), \qquad \mathrm{W} + \mathrm{S} \rightleftharpoons \mathrm{W}\cdot\mathrm{S}\ (K_W),$$
$$\mathrm{W}\cdot\mathrm{S} + \mathrm{A}\cdot\mathrm{S} \rightleftharpoons \mathrm{B}\cdot\mathrm{S} + \mathrm{S}\ (K_S), \qquad \mathrm{B}\cdot\mathrm{S} \rightleftharpoons \mathrm{B} + \mathrm{S}\ (K_B).$$

Rate laws of each step:

$$r_A = k_A\left(C_AC_v - \frac{C_{\mathrm{AS}}}{K_A}\right), \qquad r_W = k_W\left(C_WC_v - \frac{C_{\mathrm{WS}}}{K_W}\right),$$
$$r_S = k_S\left(C_{\mathrm{AS}}C_{\mathrm{WS}} - \frac{C_{\mathrm{BS}}C_v}{K_S}\right), \qquad r_B = k_B\left(\frac{C_{\mathrm{BS}}}{K_B} - C_BC_v\right).$$

Case (i): surface reaction limiting

Step 1. Fast steps: $C_{\mathrm{AS}} = K_AC_AC_v$, $C_{\mathrm{WS}} = K_WC_WC_v$, $C_{\mathrm{BS}} = K_BC_BC_v$.

Step 2. Substitute into r_S:

$$r_S = k_S\left(K_AK_WC_AC_WC_v^2 - \frac{K_BC_BC_v^2}{K_S}\right) = k_SK_AK_WC_v^2\left(C_AC_W - \frac{C_B}{K_P}\right), \quad K_P = \frac{K_SK_AK_W}{K_B}.$$

Step 3. Site balance: $C_T = C_v(1 + K_AC_A + K_WC_W + K_BC_B)$.

Step 4.

Bimolecular L–H, surface reaction limiting

$$-r'_A = \frac{k_S K_A K_W C_T^2 \left(C_A C_W - C_B/K_P\right)}{\left(1 + K_A C_A + K_W C_W + K_B C_B\right)^2}$$

The denominator is **squared** because two sites take part in the RLS (it contains C_v^2).

Case (ii): adsorption of A limiting

Step 1. Fast steps: $C_{WS} = K_W C_W C_v$, $C_{BS} = K_B C_B C_v$, and surface-reaction equilibrium:

$$C_{AS} C_{WS} = \frac{C_{BS} C_v}{K_S} \;\Rightarrow\; C_{AS} = \frac{C_{BS} C_v}{K_S C_{WS}} = \frac{K_B C_B C_v}{K_S K_W C_W} = \frac{K_A}{K_P}\frac{C_B}{C_W} C_v.$$

Step 2. RLS: $r_A = k_A C_v \left(C_A - \dfrac{C_B}{K_P C_W}\right)$.

Step 3. Site balance: $C_T = C_v \left(1 + \dfrac{K_A C_B}{K_P C_W} + K_W C_W + K_B C_B\right)$.

Step 4.

Bimolecular, adsorption of A limiting

$$-r'_A = \frac{k_A C_T \left(C_A - \dfrac{C_B}{K_P C_W}\right)}{1 + K_W C_W + K_B C_B + \dfrac{K_A C_B}{K_P C_W}}$$

Slide typo

On the slide, the last denominator term is printed as $C_B/(K_P C_W)$, and the C_{AS} line omits K_B. The correct term is $K_A C_B/(K_P C_W)$. Check: it must equal C_{AS}/C_v.

5.12 Eley–Rideal kinetics (one species reacts from the gas phase)

$\mathrm{A} + \mathrm{S} \rightleftharpoons \mathrm{A}\cdot\mathrm{S}$ (fast), $\underline{\mathrm{A}\cdot\mathrm{S} + \mathrm{B(g)} \rightleftharpoons \mathrm{C}\cdot\mathrm{S}}$ (RLS), $\mathrm{C}\cdot\mathrm{S} \rightleftharpoons \mathrm{C} + \mathrm{S}$ (fast).

Step 1. $C_{AS} = K_A P_A C_v$, $C_{CS} = K_C P_C C_v$.

Step 2. $r_S = k_S \left(C_{AS} P_B - \dfrac{C_{CS}}{K_S}\right) = k_S K_A C_v \left(P_A P_B - \dfrac{P_C}{K_P}\right)$, with $K_P = \dfrac{K_A K_S}{K_C}$.

Step 3. Site balance: $C_t = C_v(1 + K_A P_A + K_C P_C)$. **B does not appear**, because it never occupies a site.

Eley–Rideal, surface reaction limiting

$$-r'_A = \frac{k\left(P_A P_B - P_C/K_P\right)}{1 + K_A P_A + K_C P_C}, \qquad k = k_S K_A C_t.$$

Signature: B is in the numerator but not in the denominator.

5.13 Working backwards: from an observed rate law to a mechanism

Experiments often report *apparent orders*, e.g. "first order in A, zero order in B, −1 order in product C".

- A **negative order** in a species means it sits in the denominator and **dominates** it. Its adsorbed form covers most sites (e.g. $1 + K_A P_A + K_C P_C \approx K_C P_C$).
- A **zero order** in a reactant means its pressure cancels: either it appears to the same power in the numerator and the dominant denominator term, or its step is fast and not involved in the RLS.
- Write the general L–H expression, then keep only the dominant denominator term to reproduce the observed orders.

Example: A $\longrightarrow$ C, surface reaction limiting, product strongly adsorbed ($K_C P_C \gg 1 + K_A P_A$):

$$-r'_A = \frac{kP_A}{1 + K_A P_A + K_C P_C} \approx \frac{k}{K_C}\frac{P_A}{P_C} \qquad \text{(first order in A, } -1 \text{ order in C).}$$

5.14 Designing catalytic reactors with an L–H rate law

Fogler: 4th ed. §10.4

Use the same design equations as in Lectures 1–5, written per mass of catalyst W:

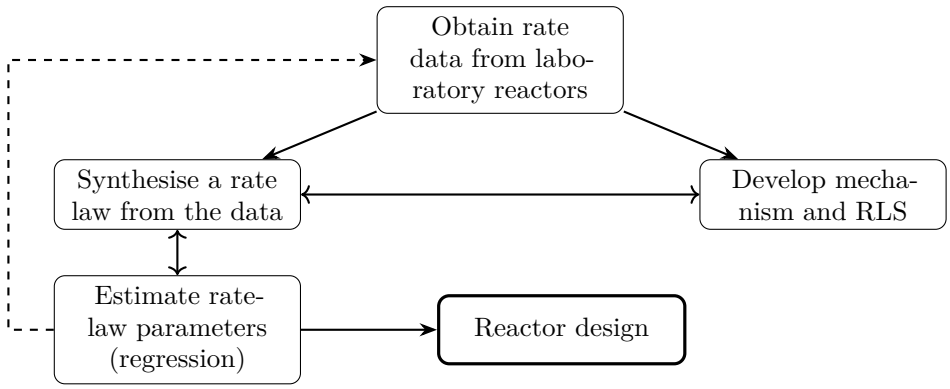
$$\text{PBR: } F_{A0}\frac{dX}{dW} = -r'_A, \qquad \text{CSTR / fluidised bed: } W = \frac{F_{A0}X}{(-r'_A)_{\text{exit}}}.$$

Stoichiometry for gases: $P_A = C_A RT = P_{A0}\dfrac{1-X}{1+\varepsilon X}\, y\, \dfrac{T_0}{T}$, and similarly $P_B = P_{A0}\dfrac{\Theta_B + (b/a)X}{1+\varepsilon X}\, y$, etc.

Worked case (PBR, analytic). $-r'_A = \dfrac{kP_A}{1 + K_A P_A}$, pure A with $\varepsilon = 0$ (A $\longrightarrow$ B), isothermal, $y = 1$:

$$W = F_{A0}\int_0^X \frac{1 + K_A P_{A0}(1-X)}{kP_{A0}(1-X)}\, dX = \frac{F_{A0}}{kP_{A0}}\left[\ln\frac{1}{1-X} + K_A P_{A0} X\right].$$

This has the same structure as the enzyme batch equation: one first-order piece plus one zero-order piece.

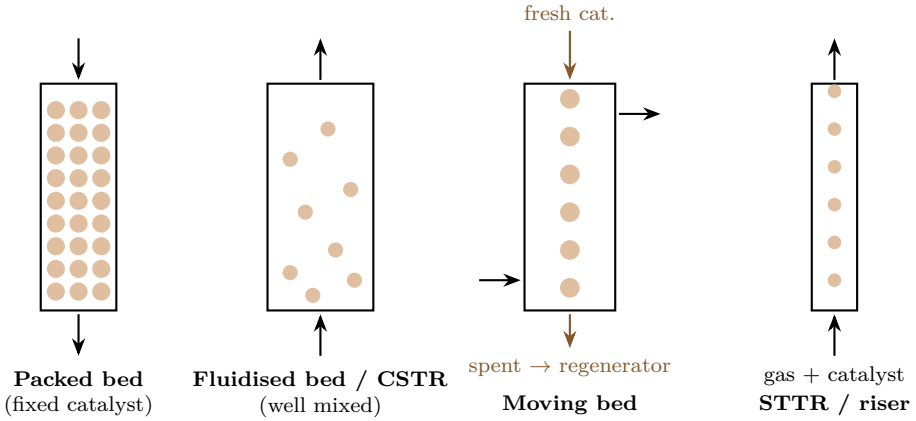


Collecting information for catalytic reactor design (Lec. 8).

6 Lecture 10: Catalytic Reactors and Catalyst Deactivation

Fogler: 4th ed. §10.7, pp. 707–730

6.1 Industrial catalytic reactors



Reactor	Description	Typical use / when to choose it
CSTR (stacked, fluidised)	Well mixed. Stacked units in series are reheated between stages for endothermic reactions (feed ∼1000°F).	Water treatment, reforming. Design: $W = F_{A0}X/(-r'_A)$.
Packed bed (PBR)	Tube filled with fixed pellets. The **highest conversion per mass of catalyst**.	Syngas, methanol, NH_3, Fischer–Tropsch, gas cleaning. Best when the catalyst decays **slowly**.
Moving bed (MBR)	Catalyst flows with the reactants, is separated, regenerated (coke burned off at ∼1400°F) and recycled.	Catalytic cracking. **Moderate** decay (hours to days).
STTR (straight-through transport)	Gas carries the catalyst straight through; fed continuously.	Gasoline from heavy oil, F–T at Sasol. **Fast** decay (seconds), so "decay disguise" is eliminated.

Catalysts and supports. Some catalysts are the active material only; most are an active phase on a support. **Zeolites** are crystalline aluminosilicates with molecular-size channels (∼1 nm), which give shape selectivity (e.g. LaY zeolite for cracking). **SBA-15** is an ordered mesoporous silica with larger pores. Examples of active phase on support: Pt–Rh/Al_2O_3 (auto exhaust), Fe/K/Al_2O_3 (NH_3), Ni/Al_2O_3 (steam reforming), Co/SiO_2 and Fe/K/Cu/SiO_2 (Fischer–Tropsch), Pt/Al_2O_3 (CO oxidation).

6.2 Separable kinetics and catalyst activity

Activity

If the kinetics are **separable**, the rate on a used catalyst is the fresh rate times an activity:

$$-r'_A = a(t) \times (-r'_A)_{\text{fresh}} = a(t)\, k(T)\, f(C_i), \qquad a(t) = \frac{-r'_A(t)}{-r'_A(t=0)}, \quad a(0) = 1.$$

The decay law is $r_d = -\dfrac{da}{dt} = p(a)\, k_d(T)\, h(C_i)$, where $p(a)$ is usually a^n and $h(C_i)$ depends

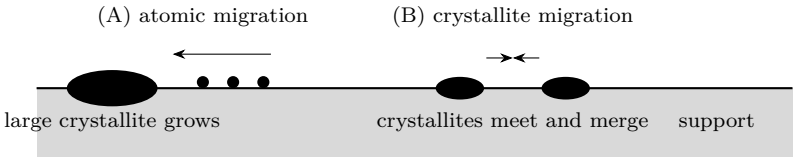
on the concentrations of poison or coke precursor. If the kinetics are **non-separable**, $-r'_A = f(\text{catalyst history}, \text{kinetics})$ and the two cannot be studied separately.

Commercial reactors keep production constant as a falls by **raising** T, which increases k (see the trajectory below).

6.3 Mechanism 1: Sintering (ageing, thermal degradation)

Fogler: 4th ed. p. 711

- Loss of active surface area after long exposure to high temperatures: crystallites grow and agglomerate, pores narrow or collapse, the support area is lost, and the surface recrystallises.
- Two models of crystallite growth: **(A) atomic migration**, where atoms leave small crystallites and join large ones, and **(B) crystallite migration**, where whole crystallites move and merge.
- Slow, **irreversible** (or hard to reverse): sintering is *prevented* rather than cured.



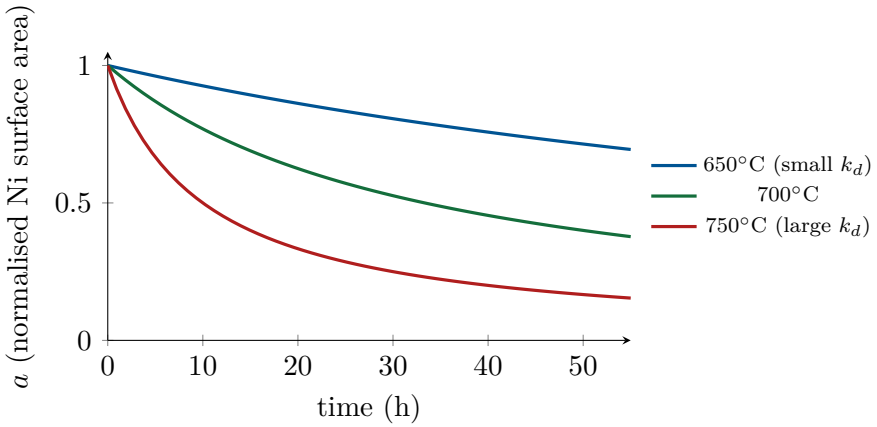
Sintering is typically second order in activity. Derivation:

$$r_d = -\frac{da}{dt} = k_d a^2 \;\Rightarrow\; \int_1^a \frac{da}{a^2} = -k_d \int_0^t dt \;\Rightarrow\; \left[-\frac{1}{a}\right]_1^a = -k_d t \;\Rightarrow\; \frac{1}{a} - 1 = k_d t.$$

Sintering (second-order decay)

$$a(t) = \frac{1}{1 + k_d t}, \qquad k_d(T) = k_d(T_0) \exp\left[\frac{E_d}{R}\left(\frac{1}{T_0} - \frac{1}{T}\right)\right].$$

Test the data by plotting $1/a$ against t: a straight line with slope k_d and intercept 1.



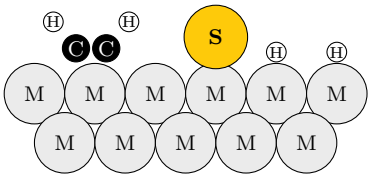
Shape of the 13.5% Ni/SiO_2 sintering data (J. Catal. 81 (1983) 131): higher T gives faster loss of area.

What changes the sintering rate: it grows exponentially with T ($E_d \approx 30$–150 kJ/mol). It depends on the atmosphere (for supported Pt the rate decreases in the order $NO > O_2 > H_2 > N_2$), on the metal

(stability in H_2: Ru > Ir ≈ Rh > Pt), on the support (Al_2O_3 > SiO_2 > carbon in stability), on promoters (Ba, Ca, Sr oxides trap atoms and slow sintering; Pb, Bi, Cl, F, S speed it up), and on pores (porous supports sinter more slowly).

6.4 Mechanism 2: Poisoning

- **Strong (irreversible) chemisorption** of a reactant, product or **impurity** on active sites. Sulfur is the classic poison in petrochemical processing.
- "Poison" has an operational meaning: O_2 is a *reactant* for ethylene oxidation on Ag, but a *poison* for ethylene hydrogenation on Ni. Poisons also change the electronic and geometric structure of the surface.



sulfur blocks sites (and neighbours) needed for ethylene hydrogenation

Poisoning kinetics. Mechanism: $A + S \rightleftharpoons A\cdot S$, $A\cdot S \rightleftharpoons B\cdot S$, $B\cdot S \rightleftharpoons B + S$, plus the poison $P + S \longrightarrow P\cdot S$ (irreversible). Then

$$-r'_A = a(t)\left[\frac{kP_A}{1 + K_A P_A + K_B P_B}\right], \qquad r_d = -\frac{da}{dt} = k_d P_P^n a^q = k'_d a^q,$$

where $k'_d = k_d P_P^n$ is constant when the impurity pressure P_P is constant. **Integrate** (for $q \neq 1$):

$$\int_1^a a^{-q}\,da = -k'_d \int_0^t dt \;\Rightarrow\; \frac{a^{1-q} - 1}{1-q} = -k'_d t \;\Rightarrow\; \boxed{a = \left[1 - (1-q)k'_d t\right]^{1/(1-q)}}$$

For $q = 1$: $a = e^{-k'_d t}$. For $q = 2$: $a = 1/(1 + k'_d t)$, the same form as sintering. For $q < 1$ the activity reaches zero in a finite time, $t = 1/[(1-q)k'_d]$.

Poisons for important catalysts (condensed from the slide tables):

Catalyst	Reaction	Poisons
Silica–alumina, zeolites	Cracking	organic bases, heavy metals
Ni, Pt, Pd	(De)hydrogenation	S, P, As, Zn, Hg, halides, Pb, NH_3, C_2H_2
Ni	Steam reforming	H_2S, As
Fe, Ru	NH_3 synthesis	O_2, H_2O, CO, S, C_2H_2
Co, Fe	Fischer–Tropsch	H_2S, COS, As, NH_3, metal carbonyls
Ag	Ethylene oxidation	C_2H_2
Pt, Pd	CO / HC oxidation	Pb, P, Zn, SO_2, Fe

By chemical type: Group VA/VIA (N, P, As, S, Se…) and Group VIIA (halogens) act through s- and p-orbitals; toxic heavy metals (As, Pb, Hg, Zn, Cu…) occupy d-orbitals and may form alloys; molecules with multiple bonds (CO, NO, HCN, benzene, acetylene) chemisorb strongly through multiple bonds and back-bonding.

6.5 Mechanism 3: Fouling and coking

- Carbonaceous deposits (coke) build up on the surface. They have two effects: **site coverage** (they block active sites) and **pore blockage** (they create diffusion resistance, and can encapsulate crystallites).
- Coking is not well understood mechanistically. Pore-blocking models are probabilistic: the probability that a site is accessible depends on the pore-size distribution.
- **Reversible**: coke can be burned off (regeneration). This is why moving beds and FCC risers have a regenerator. The burn-off is strongly exothermic, and the heat is recovered (it drives the endothermic cracking and makes steam).

Coking

Coke on the surface follows an empirical law, $C_c = At^n$ (A and n depend on the feed). The activity is then one of

$$a = \frac{1}{1 + C_c^{\,p}} = \frac{1}{1 + A^p t^{np}}, \qquad a = e^{-\alpha_1 C_c}, \qquad a = \frac{1}{1 + \alpha_2 C_c}.$$

Chemistry: CO dissociates to atomic C_α, which can be hydrogenated (CH_4), dissolve in Ni, form Ni_3C, or polymerise to less reactive C_β and then to graphitic C_c (coke). Each path has its own activation energy, so the dominant path depends on T.

Solid-state reactions that create inactive phases: $Rh_2O_3 + Al_2O_3 \longrightarrow RhAl_2O_4$ (auto catalysts), $Fe \longrightarrow FeO$ (NH_3 catalyst with >50 ppm O_2 or $H_2O/H_2 > 0.16$), $SO_3 + Al_2O_3 \longrightarrow Al_2(SO_4)_3$ (blocks pores), $Ni + Al_2O_3 \longrightarrow NiAl_2O_4$ (steam reforming), $Co + SiO_2 \longrightarrow CoO \cdot SiO_2$ (F–T), and H_2O-induced dealumination of zeolites.

6.6 Summary of decay laws

Decay order	$-da/dt$	Integrated $a(t)$	Typical cause
Zero	k_d	$1 - k_d t$	poisoning ($q = 0$)
First	$k_d a$	$e^{-k_d t}$	poisoning, some sintering
Second	$k_d a^2$	$\dfrac{1}{1 + k_d t}$	**sintering**
Order q	$k'_d a^q$	$[1 - (1-q)k'_d t]^{1/(1-q)}$	poisoning with a constant impurity
Coking	–	$\dfrac{1}{1 + A^p t^{np}}$	coke $C_c = At^n$

6.7 Using $a(t)$ in reactor problems (exam-style applications)

(1) Conversion over time (slow decay). The catalyst decays much more slowly than the gas residence time, so at each moment the reactor is at steady state with $k \to a(t)k$. For a first-order reaction with $\varepsilon = 0$ and no ΔP:

$$\text{PBR: } X(t) = 1 - \exp\left[-\frac{a(t)\,k'W}{v_0}\right], \qquad \text{CSTR (fluidised, uniform } a\text{): } X(t) = \frac{a(t)\,k'W/v_0}{1 + a(t)\,k'W/v_0}.$$

(2) Keeping conversion constant by lowering the feed. Conversion depends on the group $a\,k'W/F_{A0}$ (or $a\,k'W/v_0$), so to keep X fixed you set $F_{A0}(t) = a(t)\,F_{A0}(0)$.

(3) Temperature–time trajectory (raise T to keep the rate constant). Constant rate at constant composition needs $a(t)\,k(T) = k(T_0)$:

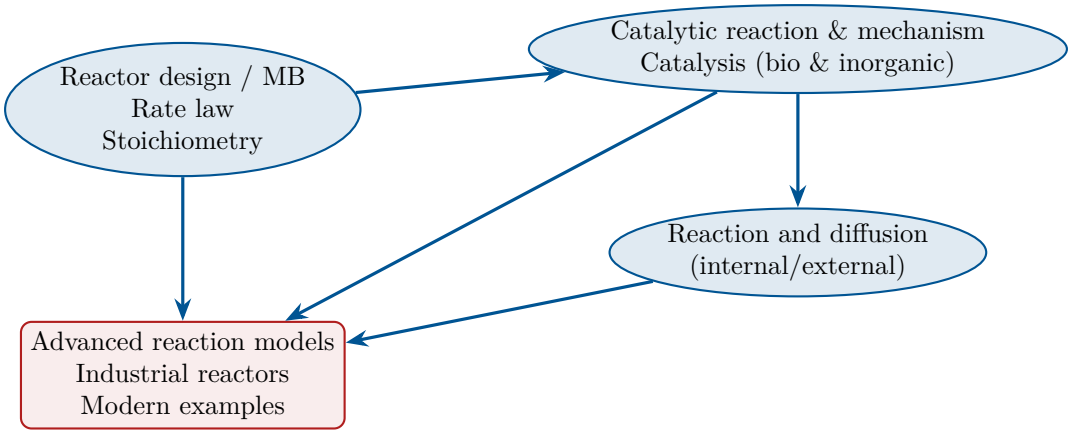
$$a(t)\exp\left[\frac{E}{R}\left(\frac{1}{T_0}-\frac{1}{T}\right)\right] = 1 \;\Rightarrow\; \frac{1}{T} = \frac{1}{T_0} + \frac{R}{E}\ln a(t).$$

With sintering, $a = 1/(1+k_d t)$, this gives $\dfrac{1}{T} = \dfrac{1}{T_0} - \dfrac{R}{E}\ln(1+k_d t)$ (ignoring the T-dependence of k_d). The real plant limit is reached when T hits a material or selectivity limit; then the catalyst is regenerated or replaced.

Common mistake

Raising T also speeds up *sintering* (k_d increases), so this strategy shortens catalyst life. If an exam problem gives E_d, you must include $k_d(T)$.

6.8 Course map



Lectures 1–5: left bubble. Lectures 6–10: top bubble. Next: diffusion.

7 Formula Summary (Lectures 6–10)

PSSH. $r_{A^*} = \sum_i r_{iA^*} = 0$. Convert each step's rate to the intermediate using $\frac{r_A}{-a} = \frac{r_B}{-b} = \frac{r_C}{c}$ (factors of 2 and $\frac{1}{2}$!).

Lindemann. $-r_A = \dfrac{k_1 k_3 C_A^2}{k_3 + k_2 C_A}$: 2nd order at low C_A, 1st order at high C_A.

Inert collider. $-r_A = \dfrac{k_1 k_3 C_A C_I}{k_3 + k_2 C_I} \to \dfrac{k_1 k_3}{k_2} C_A$.

NO oxidation. $r_{\mathrm{NO_2}} = \dfrac{k_1 k_3}{k_2} C_{\mathrm{NO}}^2 C_{\mathrm{O_2}}$, $E_{\mathrm{app}} = E_1 + E_3 - E_2$ (can be < 0).

Chain (H_2/Br_2). Initiation = termination: $2k_1 C_{\mathrm{Br_2}} = k_5 C_{\mathrm{Br}\bullet}^2$.

$$r_{\mathrm{HBr}} = \frac{2k_2\sqrt{2k_1/k_5}\, C_{\mathrm{H_2}} C_{\mathrm{Br_2}}^{3/2}}{C_{\mathrm{Br_2}} + (k_4/k_3) C_{\mathrm{HBr}}}$$

Michaelis–Menten. $-r_S = \dfrac{V_{\max} C_S}{K_M + C_S}$, $V_{\max} = k_{\mathrm{cat}} C_{E,t}$, $K_M = \dfrac{k_2 + k_3}{k_1} = S_{1/2}$.

Lineweaver–Burk. $\dfrac{1}{-r_S} = \dfrac{1}{V_{\max}} + \dfrac{K_M}{V_{\max}} \dfrac{1}{C_S}$.

Hanes–Woolf. $\dfrac{C_S}{-r_S} = \dfrac{K_M}{V_{\max}} + \dfrac{C_S}{V_{\max}}$.

Eadie–Hofstee. $-r_S = V_{\max} - K_M \dfrac{-r_S}{C_S}$.

Batch enzyme. $t = \dfrac{K_M}{V_{\max}} \ln \dfrac{1}{1-X} + \dfrac{C_{S0} X}{V_{\max}}$.

Inhibition ($\phi = 1 + C_I/K_I$):

- competitive: $\dfrac{V_{\max} C_S}{C_S + K_M \phi}$ (slope $\times\phi$)
- uncompetitive: $\dfrac{V_{\max} C_S}{K_M + C_S \phi}$ (intercept $\times\phi$)
- noncompetitive: $\dfrac{V_{\max} C_S}{(K_M + C_S)\phi}$ (both $\times\phi$)

Langmuir isotherms.

$$C_{\mathrm{AS}} = \frac{K_A P_A C_t}{1 + K_A P_A}, \qquad C_{\mathrm{AS}} = \frac{\sqrt{K_A P_A}\, C_t}{1 + \sqrt{K_A P_A}}$$

Fast step at equilibrium. $C_{\mathrm{iS}} = K_i P_i C_v$ (or $\sqrt{K_i P_i} C_v$ if dissociative).

Site balance. $C_t = C_v + \sum_i C_{\mathrm{iS}}$.

A $\rightleftharpoons$ B with $K_P = K_A K_S / K_B$:

- surface rxn RLS: $\dfrac{k(P_A - P_B/K_P)}{1 + K_A P_A + K_B P_B}$
- adsorption RLS: $\dfrac{k_A C_t (P_A - P_B/K_P)}{1 + \frac{K_A}{K_P} P_B + K_B P_B}$
- desorption RLS: $\dfrac{k(P_A - P_B/K_P)}{1 + K_A(1 + K_S) P_A}$

Dual site. $\dfrac{k(\ldots)}{(1 + \sum K_i P_i)^2}$; the initial rate has its maximum at $P_{A0} = 1/K_A$.

Eley–Rideal. $\dfrac{k(P_A P_B - P_C/K_P)}{1 + K_A P_A + K_C P_C}$ (B not in the denominator).

Catalytic design. PBR $F_{A0} \frac{dX}{dW} = -r'_A$; CSTR $W = \frac{F_{A0} X}{-r'_A}$; $P_A = P_{A0} \frac{(1-X)}{1+\varepsilon X} y \frac{T_0}{T}$.

Deactivation. $-r'_A = a(t)(-r'_A)_{\mathrm{fresh}}$; $-\frac{da}{dt} = k_d a^n$.

- sintering: $a = \dfrac{1}{1 + k_d t}$
- poisoning: $a = [1 - (1-q) k'_d t]^{1/(1-q)}$; $q = 1$: $e^{-k'_d t}$
- coking: $C_c = A t^n$, $a = \dfrac{1}{1 + A^p t^{np}}$
- T trajectory: $\frac{1}{T} = \frac{1}{T_0} + \frac{R}{E} \ln a(t)$

Arrhenius. $k(T) = k(T_0) \exp\left[\frac{E}{R}\left(\frac{1}{T_0} - \frac{1}{T}\right)\right]$; $R = 8.314$ J/(mol K)$= 1.987$ cal/(mol K).